

29050	RISK DESCRIPTION	STATUS	TREND	MANAGED (CURRENT)	TARGET (RESIDUAL)
	CHEE3007 Process Modelling and Dynamics Practicals	Live		Low	Low

RISK OWNER	RISK IDENTIFIED ON	LAST REVIEWED ON	NEXT SCHEDULED REVIEW	LOCATION CATEGORY
Naif Bilal Raja	27/06/2023	28/06/2023	28/06/2028	Facility - Laboratory

RISK FACTOR(S)	EXISTING CONTROL(S)	MANAGED (CURRENT)	PROPOSED CONTROL(S)	TREATMENT OWNER	DUE DATE	TARGET (RESIDUAL)
Ethylene glycol is a flammable liquid with a low flash point and can pose a significant risk if exposed. It is toxic if ingested or inhaled, and can cause skin and eye irritation. Direct contact or inhalation of ethylene glycol vapors can lead to health complications.	Control: 1. Ventillation system in place to eliminate buildup of hazardous vapours 2. Pressure relief system and emergency shutdown in place to avoid over pressurisation. 3. Electrical cables and wires undergo thorough testing and tagging procedures performed exclusively by certified UQ electricians to mitigate the risk of ignition resulting from potential short circuits.	Low				Low
	Control: 1. Implementation standard operating procedures (SOPs) for the safe handling of chemicals and operation of the distillation column. 2. All personnel engaged in pilot plant operations receive thorough training, focusing on hazard awareness, emergency procedures, and proper utilization of personal protective equipment (PPE).					
	Control: 1.Mandatory use of appropriate PPE, including gloves, goggles, lab coats, and/or respiratory protection, when working with or in proximity to hazardous substances. 2. Emergency eyewash stations, safety showers, and spill response kits are provided					

	and strategically placed in readily accessible locations throughout the facility.					
The distillation column operation poses risks of fires and burns due to high temperatures and pressures. Accessing the feed tray via a staircase introduces potential hazards of trips and falls. Additionally, the use of water and ethylene glycol in the process presents a risk of slips caused by spills or leaks, particularly in the event of equipment failure.	Control: 1. Physical shielding is implemented to separate the reboiler system, which operates at the highest temperature within the distillation column. Control: 1. Pressure relief system and emergency shutdown in place to avoid over pressurisation. 2. Electrical cables and wires undergo thorough testing and tagging procedures performed exclusively by certified UQ electricians to mitigate the risk of ignition resulting from potential short circuits. 3. Electrical cables and wires undergo thorough testing and tagging procedures performed exclusively by certified UQ electricians to mitigate the risk of ignition resulting from potential short circuits. 4. The facility includes a sprinkler system to mitigate the risk of fire. 5. Regular inspection are held to maintain the equipment to avoid leaks or failures. Control: 1. To mitigate slips, trips, and falls, it is mandated that a maximum of two individuals (Max 135 kg) use the staircase at a time. 2. Implementation standard operating procedures (SOPs) for the safe handling of chemicals and operation of the distillation column. 3. Incident and near-miss reporting via UQSafe.	Low				Low

4. Training provided for personnel in pilot plant operations, emphasizing hazard awareness, emergency procedures, and PPE usage.

Control: 1.Mandatory use of appropriate PPE, including gloves, goggles, lab coats, and closed shoes, while working on the distillation column.

ATTACHMENTS

ethylene glycol_2022_December_23_AUS_EN.pdf